Mapping Race and Housing Market Estimates in NYC
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This analysis intends to examine the possibility of a visible association between the racial makeup of an area and the amount of housing purchased by investors. Specifically, this report investigates a link between non-residential purchases and the percentage of people in an area who are native Hawaiian and other Pacific Islanders, American Indian and Alaska natives, some other race alone, or two or more races (called the other races group. By comparing these two variables, we will be able to detect patterns of segregation and gentrification throughout the city. The objective of this report is to identify how the other races group is being affected by outsider purchasing of housing.

To start the analysis, we focused on the background layer, which uses the Location quotient (LQ) for the 'other races' group population in NYC. The LQ is a measure that identifies population concentration in different areas, and using this measure three main clusters were identified: two in the county of Queens and one on the south-east side of the Bronx. For this layer, the manual classification scheme is used, with a focus on splitting data above and below one because this is the value where the population concentrations for the areas are equal. This method gives us the liberty to control the lower and upper part of our data range, and can also allow us to isolate extreme values. Because we are working with a ratio a divergent color ramp was implemented. A neutral color is the equilibrium point, with brown and cyan as the opposite ends. This highlights the concentration of the 'other groups' population and contrasts areas of higher concentration with lower concentration. The color selection does not have any intended meaning and was chosen to create a background that contrasts with our primary analysis layer. These colors are set to be fully opaque with faded tract boundaries for a cleaner map, allowing the patterns to be spotted more easily.

The primary layer focuses on the Redfin data, specifically the percentage of homes purchased by investors. For this layer we used manual classification scheme, which allows us to split the percentage under an equal counts scheme for the first 4 breaks (14.3, 21.4, 33.3 and 90), and then add an extra break (90-100) to spot outliers. This decision was made with our analytical objective in mind; because as mentioned in the thematic mapping lecture, equal count scheme helps detect patterns. Three main clusters can be spotted in the primary layer: one in south New York County, one in north Kings County, and one in west Bronx. The color ramp is an ascending scheme based on red, which was chosen for contrast with the background layer and because it is the brand color for Redfin. The color ramp is transparent enough so information from the background layer is still visible. Besides the color, ascending graduated size circular symbols help pinpoint the areas with higher percentages of investor purchasing, making the pattern sharper.

The west Bronx is the only area where one of the clusters in our primary layer significantly superimposes over a cluster in the background layer. A high percentage of homes purchased by investors in neighborhoods like Mount Hope, High Bridge, and East Tremont matches a high ratio of other races population. While the area of the cluster in the primary layer doesn't cover the whole background cluster, it does have significant overlap that is not present in the other two background clusters. Although this area is a visually dense spot for both layers, the transparency of the symbols in the primary layer lets the background layer and the neighborhood labels remain legible.

By analyzing the percentage of houses purchased by investors within an area of high concentration of the other races group, one area of significant overlap was identified in a specific region with in one of the population clusters. Based solely on this result, it is hard to tell if the other races group is experiencing displacement in the area of interest. Further study would need to be done to determine the specific impacts, but from the map generated we can observe that high purchasing of homes by investors is happening in a limited area of high other races concentration.
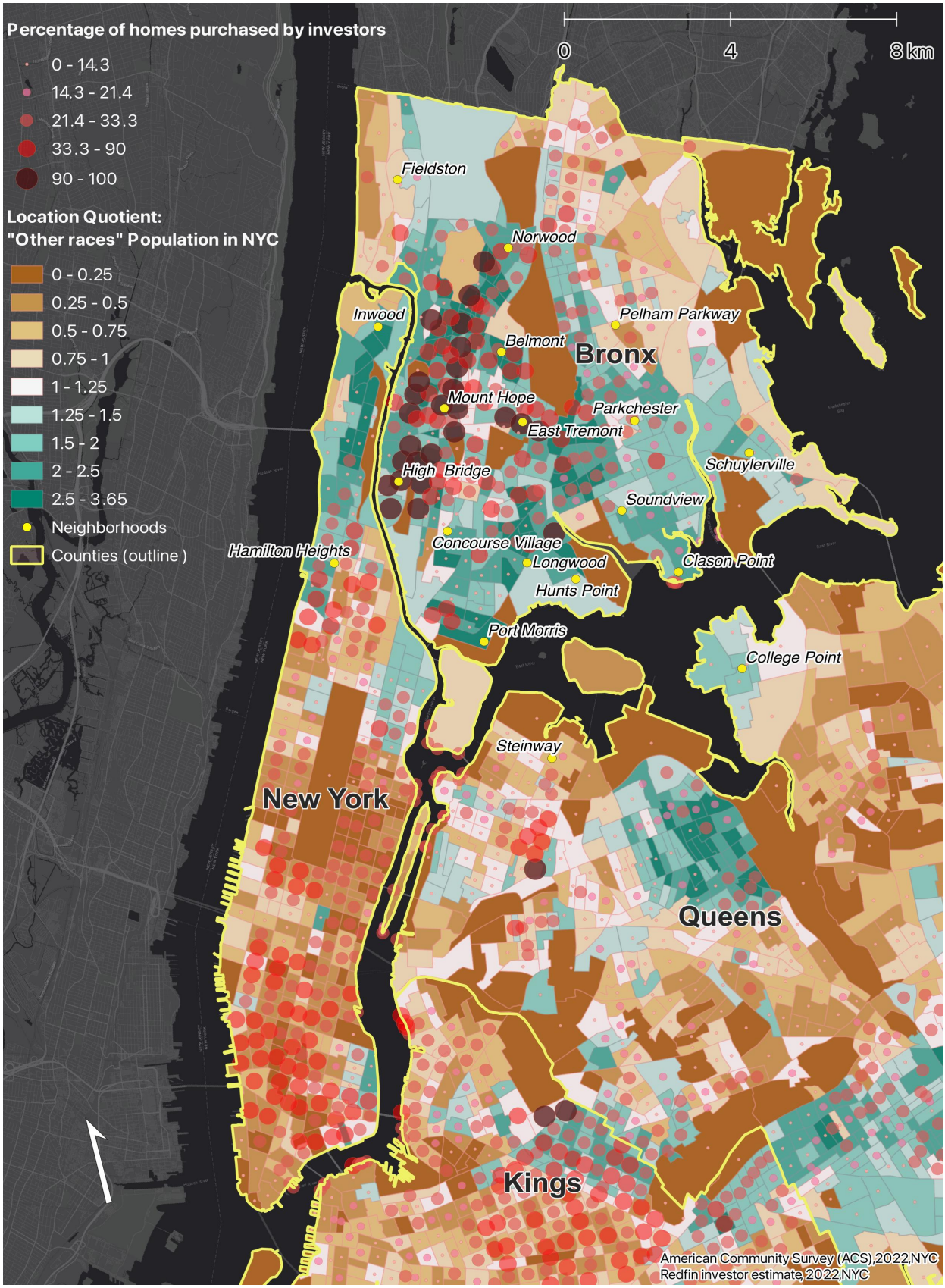
Figure 1, "Investor Housing Purchases vs Other Races Demographics: Identifying Patterns of Segregation and Gentrification"